Shivanshu Ojha

Tbilisi, Georgia • ojhashiv@msu.edu • linkedin.com/in/ojhashiv • GitHub

EDUCATION

Michigan State University, East Lansing, MI May 2026
Bachelor of Science, *Electrical Engineering*
Capstone: *Multi-objective optimization of AlGaN/AlN epitaxial stacks using genetic algorithms to reduce reverse-bias leakage in ultra-wide bandgap semiconductors using for power and RF application.*
Advisor: *Dr. Nicholas C. Miller*
GPA: 3.71/4.00

SKILLS

Programming Languages: Python, C/C++, MATLAB, SQL, Fortran
Design: SOLIDWORKS, COMSOL, Synopsys Sentaurus, Custom Device Simulator, ADS, Cadence OrCAD and Virtuoso
Physical: Oscilloscopes, VNA, Class 100 Cleanroom Fabrication, Schlieren & Fourier Optics

EXPERIENCE

Electromagnetics Research Group, East Lansing, MI Jan. 2026 to Oct. 2026
Research Assistant

- Developed optimization methodologies for coordinated distributed antenna arrays, integrating evolutionary computation such as genetic and memetic algorithms, local-search heuristics, and large-scale numerical experimentation to improve Fourier-domain sampling efficiency.
- Evaluated and benchmarked series and inset-fed as well as travelling-wave antenna topologies, optimizing for directivity and beamforming performance for incoherent interferometry at 75 GHz on Ansys HFSS. Architected a custom 75 GHz integrated receiver chain incorporating PIN diode switches, bandpass filters, LNAs, mixers and ADCs to enable high-resolution spatial imaging.

RF Design Center, East Lansing, MI Jan. 2025 – Jan. 2026
Research Assistant

- Simulated a GaN/AlGaN HEMT device structure from Tunga et al. (JAP, 2022); simulated trapping effects, 2DEG formation and other HEMT phenomena such as interface traps on Synopsys Sentaurus; developed an internal guide to Synopsys software for use by future undergraduate researchers.
- Developed a full one-dimensional drift-diffusion solver coupling Poisson and carrier continuity equations using Scharfetter Gummel discretization; validated voltage sweeps under diverse boundary conditions, focusing on PIN and PN diodes, solver titled, *SpartyCAD*. See GitHub link for further details.
- Built a multi-objective optimization framework using *pymoo* NSGA-II for linearly graded AlGaN epitaxial stacks utilizing specialized TCAD-surrogate modeling, enabling exploration of design spaces of 10,000 contact candidates within ten minutes to improve tunneling field emission using similar mechanisms.
- Modelled the epitaxial stack from Chen et al., (IEEE TED 2025, UW-Madison), replicating the reverse-graded n-Al(x)Ga(1-x)N (20nm)/Al(x)Ga(1-x)N (22nm) diode beneath source/drain contacts; and implemented charge transport models to analyze reverse-bias leakage. Applied a physics-informed optimization technique to design and simulate new reverse-graded $Al_xGa_{1-x}N$ stacks to reduce contact resistance.

Bayer, Pittsburgh, PA May 2025 – Aug. 2025
Device Early Research and Development Intern

- Engineered custom RF oscillator and amplifiers to drive an intensity-based sensing platform for real-time air and fluid detection in MRI contrast injectors using OrCAD and packaged for biomedical radiation standards for MRI injectors.
- Designed and fabricated custom STM32-based integration boards to generate bio-inspired ECG signals for mixed-signal testing and real-time analog control of flow rates to a test a circulation phantom.
- Developed a modular RF testbench with impedance-matched 2-port networks using ADS, enabling rapid prototyping and device characterization under IEC 60601-1 and EMI compliance.
- Built custom 2D FDTD and Schlieren-based simulation and optical bench tools to optimize light coupling through fluids, lenses, and other multilayer sensor stacks, reducing optical power loss by 15% and enabling first-in-company high-frequency image processing algorithm development to visualize dual-flow injections.

MSU Institute for Quantitative Health Science and Engineering, East Lansing, MI Dec. 2023 – Jan. 2025
Undergraduate Research Assistant

- Conducted research as an Electrical Engineering Research Assistant, focusing on designing and manufacturing a microfluidic device for biomedical applications, adhering to lab SOPs for precise designs.

- Leveraged Simulink for analog controls, optimized manufacturing using SPICE, COMSOL, and MATLAB, and developed skills in using power tools and 3D printers with SOLIDWORKS to build the electro-microfluidic device.

PUBLICATIONS

[1] S. K. Shiran, J. M. Merlo, **S. Ojha**, J. R. Colon-Berrios, J. A. Nanzer, ***Towards Fourier-domain imaging using wirelessly coordinated distributed antenna arrays***, SPIE CI/CS: Computational Imaging, Compressed Sensing and Sensors (Security + Defense), presented at Edinburgh, United Kingdom, in September 2026

[2] S. K. Shiran, J. M. Merlo, **S. Ojha**, J. R. Colon-Berrios, J.B. Lancaster, J. A. Nanzer, *Fourier Domain Synthesis Imaging Using a Wirelessly Coordinated Distributed Antenna Array*, arXiv:2609.19562. Submitted to *IEEE Transactions on Microwave Theory and Techniques*.

PROJECTS

Passive Electromagnetic Devices

- For these projects, we designed a patch antenna operating at 5.8 GHz, as well as 0.1 dB ripple transmission-line based filters designed for 1-3 GHz, as well as a wideband substrate integrated waveguide and a SIW power-divider operating from 8 to 10 GHz.

Eight-Bit Microprocessor Datapath

- Led 8x8 SRAM and dynamic circuit design for a custom CMOS-based 8-bit data path using Cadence Virtuoso. Achieved 0.0112 tx/μm² transistor density and SRAM timings of 1.11 ns (write), 2.86 ns (read), and 300–900 ps 8-bit domino decoder, as well as <10ns for full data path on the AMIC5N process. Received Outstanding Project Award.

Silicon Device Fabrication

- Fabricated devices on a p-type Si wafer, using PVD, PECVD, RCA processes and lithography. Measured device performance post-fabrication; simulated process flow and device physics on Synopsys Sentaurus, comparing experimental results to analytical calculations; analyzed a fabricated Si resistor with sheet resistance equal to 366 (Ω/□).

Power Amplifier Design

- Designed a power amplifier using a GaAs pHEMT from 0.8 to 1.2 GHz with 400 MHz bandwidth. Used discrete parts and soldered all together on a layout generated from Keysight ADS. Performed mixed-element impedance matching for a RF Circuits class.

TEACHING EXPERIENCES

MSU College of Engineering, East Lansing, MI — Aug. 2024 – May 2026

Undergraduate Learning Assistant, Electrical and Computer Engineering

- Graded and provided feedback on assignments for over 300 students across ECE 201, ECE 377, ECE 410. Assisted numerous faculty members (Drs. Kalyanmoy Deb, Nicholas Miller, Shannon Nicley) in administering coursework.
- Supported student learning through help room guidance, fostering deeper conceptual understanding of core EE subjects such as circuits, semiconductor devices and VLSI Design.

MSU College of Natural Science, East Lansing, MI — Jul. 2023 – Dec. 2023

Undergraduate Learning Assistant, Department of Mathematics

- Facilitated activities three times a week at the Math Learning Center to enhance students' understanding of mathematical material as well as teaching two days a week. Improved student comprehension and academic performance, by a 20% increase in grades.
- Devised study plans and advised five students each month, monitored exams and quizzes for 200 students, and provided targeted support and explanations

GRANTS APPLIED FOR

- ***NASA OpenSkies****,* $9000 for CRACKS: *a partnered venture between Michigan's Department of Transportation for drone-based NDE to detect faults based on InSAR*, for DRIFT.

EXTRACURRICULAR AND VOLUNTEER EXPERIENCES

DRIFT, East Lansing, MI — August 2026 – Present

Technical Advisor

- Worked on advising students and former 2x NASA USRC winners. Also worked on securing funding from NASA Blue Skies grants for development of UAV and drones based on InSAR. Currently working on reconfigurable mesh network design using neural network integrating with GNSS for positional correction.

QuantZ, East Lansing, MI — May 2024 – January 2026

Vice President and Interim Treasurer

- Co-founded and led a 20-member interdisciplinary team of engineers and business students to develop MSU's first digital signal processing unit tailored for high-frequency trading. Sponsored by Jane Street (a quantitative trading firm and liquidity provider).
- Directed mathematical modelling efforts in theoretical stochastic processes and chaos theory to enhance predictive analysis in market behaviour, applying Python for simulation.

Michigan State Rocketry Team, East Lansing, MI Sep. 2023 – May 2024
Electrical Division Antenna Lead

- Worked on developing stable flight computers, parachute systems, flight altimeters, in-flight cameras, and native communication systems within the 900 MHz range for a club project.
- Designed and implemented systems to meet stability and performance specifications, consulted with the mechanical department on aerodynamic fins for antennas, and utilized Raspberry Pi, OrCAD, and SPICE for PCB design and circuit simulations.

AWARDS

- International Tuition Grant, $35K/year, Michigan State University, awarded 2022 through 2026
- Presidential Study Abroad Grant, $5,000, Michigan State University, awarded 2022 through 2026
- Outstanding Project Award, $500, Michigan State University, awarded 2025
- Dean's List, Spring 2023 – Spring 2026

REFERENCES

Dr. Edward Gebara, Associate Professor, Michigan State University
Dr. William Barone, Assistant Director, Bayer Radiology
Dr. Kalyanmoy Deb, Shelby Endowed Professor, University of Alabama